

Lecture 18 Notes: Risk, Return, and the CAPM Framework

Corporate Finance

I. Banks and Foreign Investment Restrictions

Key Points

- Banks have a deep connection with the economy due to changes in interest rates, resulting in **higher systematic risk** compared to typical companies.
- Due to restrictions on the movement of foreign currency and dollars, individual investors generally **cannot invest directly in foreign stocks**.
- Companies can invest abroad through mechanisms like **American Depositary Receipts (ADRs)**.
- The government will question significant movements of dollars because the current regime is not fully flexible regarding capital controls.

II. Sovereign Debt and Default Risks

Key Points

- Pakistan paid back approximately \$1.5 to \$1.6 billion on its bonds last week (reported by BBC).
- A country will **not typically default** on local currency or local bonds.
- Default generally occurs on **foreign obligations** (e.g., Sri Lanka).

III. Quantifying Risk: Standard Deviation vs. Beta

Two Securities Example

Security	Standard Deviation	Beta
Security A	40%	0.5
Security B	20%	1.5

Interpretation

Security A	Security B
Higher total risk (40% std dev)	Lower total risk (20% std dev)
Lower systematic risk (Beta = 0.5)	Higher systematic risk (Beta = 1.5)
Most risk is unsystematic	Most risk is systematic
Risk can be diversified away	Risk cannot be eliminated

Key Insight

- **Standard Deviation** = Measure of **total risk** (systematic + unsystematic)
- **Beta** = Measure of **only systematic risk** (market risk)
- Higher Beta $\rightarrow$ Higher systematic risk $\rightarrow$ Higher required return
- Unsystematic risk can be diversified away; the market does not reward it

IV. The Risk-Free Rate

Definition

The **risk-free rate** (r_{RF}) is the return an investor receives for taking no risk. For example, investing in Treasury bills.

Key Principles:

- The starting point for any investment return should be the risk-free rate.
- If an investor puts all money into a risk-free asset, they get the risk-free rate with zero risk.
- The market only rewards investors for risk that **cannot be eliminated** (systematic risk).
- The market does **NOT** reward unsystematic risk because investors can eliminate it through diversification.

V. Portfolio Combinations: Risky Asset + Risk-Free Asset

Example Assumptions

$$\begin{aligned}\text{Risk-free rate } (r_{RF}) &= 8\% \\ \text{Risky Asset A expected return} &= 20\% \\ \text{Risky Asset A Beta } (\beta_A) &= 1.6 \\ \text{Risk-free asset Beta } (\beta_{RF}) &= 0\end{aligned}$$

Scenario 1: 25% in Asset A, 75% in Risk-Free Asset

Portfolio weights: $w_A = 0.25$, $w_{RF} = 0.75$

Portfolio Expected Return:

$$E(R_p) = w_A \times E(R_A) + w_{RF} \times r_{RF}$$

$$E(R_p) = 0.25 \times 20\% + 0.75 \times 8\% = 5\% + 6\% = 11\%$$

Portfolio Beta:

$$\beta_p = w_A \times \beta_A + w_{RF} \times \beta_{RF}$$

$$\beta_p = 0.25 \times 1.6 + 0.75 \times 0 = 0.4$$

Scenario 2: Borrowing to Invest More Than 100%

An investor can invest **more than 100%** of initial capital by borrowing at the risk-free rate.

Borrowing Example

Initial capital = \$100

Borrow \$50 at risk-free rate (8%)

Total invested in Asset A = \$150 (150% of initial capital)

Portfolio weights: $w_A = 1.5$, $w_{RF} = -0.5$ (negative = borrowing)

Portfolio Expected Return:

$$E(R_p) = 1.5 \times 20\% + (-0.5) \times 8\% = 30\% - 4\% = 26\%$$

Portfolio Beta:

$$\beta_p = 1.5 \times 1.6 + (-0.5) \times 0 = 2.4$$

VI. The Expected Return-Beta Relationship

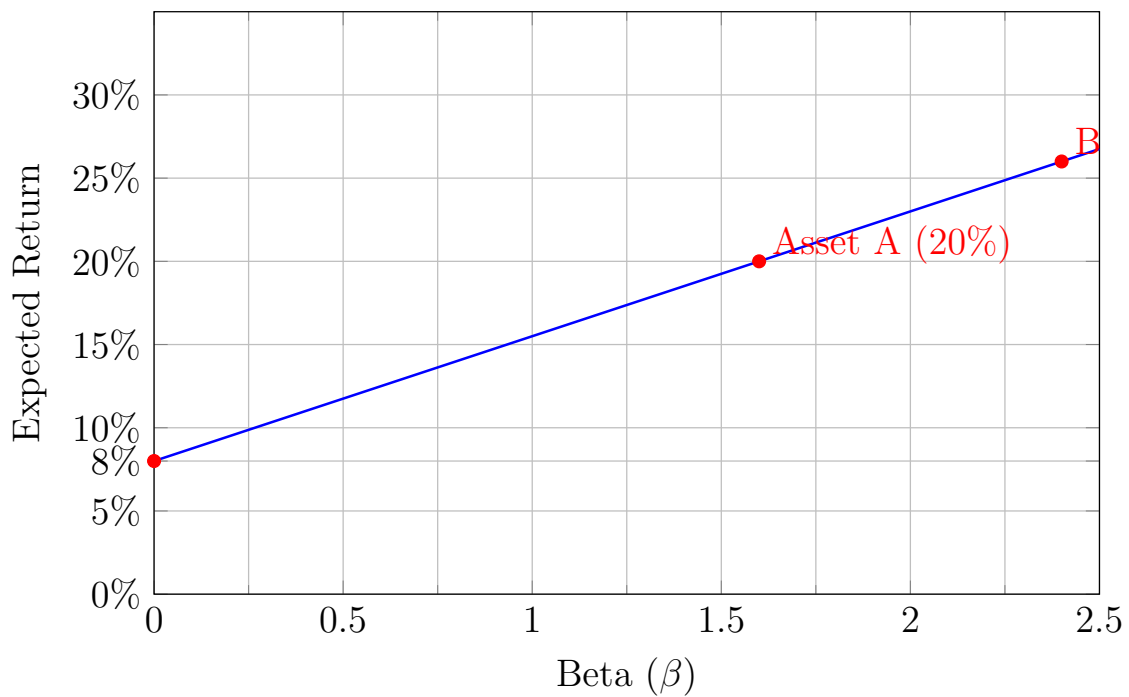
When plotting various combinations of risky and risk-free assets, a clear pattern emerges:

Key Relationship

As Beta increases, Expected Return increases.

Graph Characteristics

- The relationship forms a **straight line with a positive slope**.
- The starting point (where Beta = 0) is the **risk-free rate** (intercept).
- This line is the foundation of the **Capital Asset Pricing Model (CAPM)**.



VII. Calculating the Slope (Risk Premium)

Using the example points:

Point 1: ($\beta = 0$, $E(R) = 8\%$) Point 2: ($\beta = 1.6$, $E(R) = 20\%$)

$$\text{Rise} = 20\% - 8\% = 12\%$$

$$\text{Run} = 1.6 - 0 = 1.6$$

$$\text{Slope} = \frac{12\%}{1.6} = 7.5\%$$

Interpretation of Slope

The slope (7.5%) represents the **market risk premium**:

- For every 1 unit of systematic risk (Beta), the investor should receive a 7.5% premium over the risk-free rate.
- This is the additional return required for bearing systematic risk.

VIII. The CAPM Expected Return Equation

Expected Return Formula

$$E(R_i) = r_{RF} + \beta_i \times (E(R_m) - r_{RF})$$

Where:

- $E(R_i)$ = Expected return on security i
- r_{RF} = Risk-free rate (intercept)
- β_i = Beta of security i (systematic risk measure)
- $E(R_m) - r_{RF}$ = Market risk premium (slope)

From our example:

$$E(R_i) = 8\% + \beta_i \times 7.5\%$$

Verification for Asset A ($\beta = 1.6$)

$$E(R_A) = 8\% + 1.6 \times 7.5\% = 8\% + 12\% = 20\% \quad (\text{Matches given})$$

IX. Summary of Key Concepts

Lecture Summary

1. **Total Risk** (Standard Deviation) = Systematic Risk + Unsystematic Risk
2. **Systematic Risk** (Beta) cannot be diversified away
3. **Unsystematic Risk** can be eliminated through diversification
4. The market only rewards investors for taking **systematic risk**
5. The risk-free rate is the starting point (intercept) for expected returns
6. Investors can borrow at the risk-free rate to invest more than 100% in risky assets
7. The Expected Return-Beta relationship is a **straight line** with positive slope
8. The slope equals the **market risk premium**
9. CAPM: $E(R_i) = r_{RF} + \beta_i \times (E(R_m) - r_{RF})$

X. Closing Logistics

- Upcoming midterm exams
- Students are advised to be in "better condition" for the next class following exams

Formula Summary

$$\text{Portfolio Expected Return : } E(R_p) = \sum w_i E(R_i) \quad (1)$$

$$\text{Portfolio Beta : } \beta_p = \sum w_i \beta_i \quad (2)$$

$$\text{CAPM : } E(R_i) = r_{RF} + \beta_i (E(R_m) - r_{RF}) \quad (3)$$

$$\text{Risk Premium : } RP = E(R_m) - r_{RF} \quad (4)$$

$$\text{Slope of SML : } \text{Slope} = \frac{E(R_i) - r_{RF}}{\beta_i} \quad (5)$$